Observer Disagreement, Predictive Scale, and a Stable Component-Change Law in Conway's Game of Life

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(Dated: April 2026)

Different coarse-grained descriptions of the same deterministic system can yield different measured changes. Using a component-count fine observer and an occupied-block coarse observer, we establish three empirical regularities in Conway's Game of Life. Early component net change predicts the full-run normalized narrative gap between observers with partial correlation $r = -0.848$ (95% bootstrap CI $[-0.862, -0.830]$, $N = 1000$), sharpening after density control. The predictive scale that best forecasts a future observable depends on which observable: time-averaged live-cell count peaks at an intermediate scale ($B = 2$) while occupied-block count peaks at its own native scale ($B = 8$), with non-overlapping bootstrap confidence intervals. A linear law linking embedded isolated cells to future component-level decline is stable across six size-by-density conditions (mean slope ≈ -1.52, CV ≈ 0.09). Two stronger mechanism claims—global mediation through component net change, and residual-coupling beyond death and local-neighbourhood dynamics—do not survive their pre-specified tests. All results are established computationally within GoL; generalisation beyond this system remains an open empirical question.

Keywords: Conway's Game of Life, cellular automata, coarse-graining, observer disagreement, predictive scale, component counting, empirical scaling law, causal emergence

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I. INTRODUCTION

Different descriptions of the same dynamical system can yield different accounts of change. In Conway's Game of Life[3,12], one natural observer counts 4-connected components of live cells—each glider is one object, each still life is one object—while another counts occupied 8×8 spatial blocks. These are not straightforward approximations of a single underlying quantity. Components and blocks define different effective objects, with different characteristic lifetimes and different event statistics, and when they disagree on whether the system underwent net creation or net destruction over a given interval, neither is wrong. They track different observables.

This raises three concrete questions. First: is the disagreement between observers predictable from early dynamics alone, before the two counts have diverged significantly? Second: does the observation scale that best forecasts a future quantity depend on which quantity one is forecasting? Third: does a simple local structural count—the number of cells that are alive but 4-disconnected from all neighbours while touching at least one diagonal—predict future component-level change with a stable coefficient across different system sizes and density regimes?

We answer all three affirmatively in GoL, and we also test two stronger interpretations—that embedded isolated cells globally mediate the path from early structure to observer disagreement, and that the slope law survives as a positive residual over a dynamic local null—neither of which holds up. All analyses use a single fine observer definition throughout: 4-connected component count on the periodic torus.

a. Relation to prior work. Hoel et al.[4] characterise causal emergence as conditions under which a macro-scale description carries more effective information than the micro description; the information-integration framework from which this draws originates with Tononi et al.[10]. The present work is complementary but narrower, measuring the predictability of a signed net-change difference rather than information content. Israeli and Goldenfeld[5] showed that some cellular automata admit exact coarse-grained descriptions while others are computationally irreducible; Wolfram[12] established universality and complexity properties across rule spaces, and Langton[6] showed that GoL-like rules operate near a phase transition supporting long-lived structured computation. Our result is consistent with the irreducible case, in which the two observers compute incommensurable quantities. Shalizi and Moore[9]

argue that the success of coarse-graining depends on which features of the micro-state the macro-description preserves; the target-relative predictive structure found here is a concrete instance of that dependence. Crutchfield and Young[1] formalise macro-state identification through the epsilon-machine framework, a computational-mechanics approach complementary to the correlation-based analysis used here; Schreiber[8] operationalise cross-scale information flow through transfer entropy. Rupe and Crutchfield[7] identify local causal states in GoL that capture spatially localised computational structure; the embedded isolated cells studied here are a simpler structural class defined by connectivity rather than causal equivalence. Vichniac[11] established cellular automata as substrates for simulating physical systems, providing broader context for treating GoL as a model dynamical system. Flack[2] discusses coarse-graining as a mechanism for effective description in complex systems.

II. METHODS

A. Model dynamics and observers

Let $\mathcal{G}$ be an $N \times N$ periodic grid evolving under GoL (B3/S23) on the torus. The **fine observer** F counts the number of 4-connected components of live cells at each step, with periodicity enforced by breadth-first search across opposite edges. (`scipy.ndimage.label` operates on a flat grid and would miscount components that span opposite edges; it is not used here.) The **coarse observer** C counts occupied $B \times B$ non-overlapping blocks, where a block is occupied if any cell within it is alive; default $B = 8$. We denote these counts c_t^F and c_t^C at step t. Throughout, an observer is defined operationally as a fixed mapping from the grid state to a scalar count.

B. Derived quantities

For a time window $[a, b]$,

$$\mathrm{net}_F[a,b] = c_b^F - c_a^F, \qquad \mathrm{net}_C[a,b] = c_b^C - c_a^C, \tag{1}$$

$$\rho_F[a,b] = \mathrm{net}_F[a,b]/\max(c_a^F, 1), \qquad \rho_C[a,b] = \mathrm{net}_C[a,b]/\max(c_a^C, 1). \tag{2}$$

The **narrative gap** $G[a,b] = \rho_C[a,b] - \rho_F[a,b]$ is positive when the coarse observer records more net creation and negative when the fine observer does. The **early imbalance** $\Delta_{\mathrm{early}} =$

TABLE I. Experimental design. Section references indicate where each ensemble's results are reported.

Ensemble	Grid	N	T	Seed	Main statistic
Observer-disagreement	64^2	1000	250	20260325	Pearson/partial r
Predictive-scale	128^2	500	250	20260326	Ridge CV R^2
Component-change law	$64^2/128^2$	300–400	250	20260328+k	OLS slope

$\mathrm{net}_F[0, t_{\mathrm{early}}]$ is the fine-scale component net change over the first t_{early} steps and serves as the predictor in Section III.

C. Cell-level structural classifier

The mechanism tests in Section VI use a classifier that operates on individual cells, not on components. An **embedded** cell is alive with no 4-connected live neighbours but at least one 8-connected live neighbour. An **alone** cell is alive with no 8-connected live neighbours. These are cell-level structural categories; the observer variables above are component-level and block-level counts. The two levels are not interchangeable.

D. Experimental design and statistical analysis

We analyse three simulation ensembles; their parameters appear in Table I. Each world is constructed from a per-world integer seed derived from a global seed, so any world can be exactly regenerated from its stored seed and density. The predictive-scale analysis uses Ridge regression ($\alpha = 1.0$) with standardised inputs and 5-fold cross-validated R^2. Decision thresholds for the primary comparisons were fixed before analysis. No hyperparameter tuning is performed across conditions.

III. PREDICTABLE OBSERVER DISAGREEMENT

In a 1000-world ensemble on 64×64 GoL, initial densities drawn uniformly from $[0.03, 0.58]$, we measure whether Δ_{early}—the net change in component count over the first 50 steps—predicts the full-run normalized narrative gap $G[0, 250]$. Because initial density can

TABLE II. Key statistics for the observer-disagreement analysis ($N = 1000$, 64×64 GoL, $t_{\text{early}} = 50$, $B_{\text{coarse}} = 8$).

Statistic	Value	
$r(\Delta_{\text{early}}, G)$	-0.765	$p = 1.4 \times 10^{-192}$
OLS slope	-1.07×10^{-3}	95% CI $[-1.12, -1.01] \times 10^{-3}$
OLS intercept	0.076	
$r(\Delta_{\text{early}}, \text{net}_C)$	-0.097	$p = 0.002$
Partial $r(\Delta_{\text{early}}, G \mid \rho)$	-0.848	
Bootstrap 95% CI (partial r)	$[-0.862, -0.830]$	world-row resampling
Replication (seed 20260326)	-0.835	consistent

drive both variables—denser worlds tend to have more negative Δ_{early} and more positive G—we compute the partial correlation $r(\Delta_{\text{early}}, G \mid \rho)$, residualising both variables on initial density, and we construct a 95% confidence interval by bootstrapping over world rows (1000 resamples, with the world as the resampling unit).

Table II reports the key statistics. The raw Pearson correlation is $r = -0.765$ ($p = 1.4 \times 10^{-192}$). The partial correlation is $r = -0.848$, with a 95% bootstrap CI of $[-0.862, -0.830]$ that excludes zero by a wide margin. A replication on a second global seed gives partial $r = -0.835$, confirming the result is not an artefact of the particular seed.

The partial correlation exceeds the raw correlation because density adds scatter to the raw relationship. Denser worlds have more negative Δ_{early} and more positive G, so removing the shared density channel tightens rather than loosens the association. Within any density stratum the component net-change signal aligns more closely with the later observer gap than the raw cross-density scatter suggests.

The independence check $r(\Delta_{\text{early}}, \text{net}_C) = -0.097$ ($p = 0.002$) confirms that early fine-scale dynamics carry genuine, if modest, predictive content about coarse-scale net change over and above the density channel. The result therefore does not reduce to a density confound.

Figure 2 shows illustrative cumulative net-change traces for two representative worlds: one with positive gap in which the fine observer records large net destruction while the coarse observer remains near zero, and one with negative gap showing the opposite pattern.

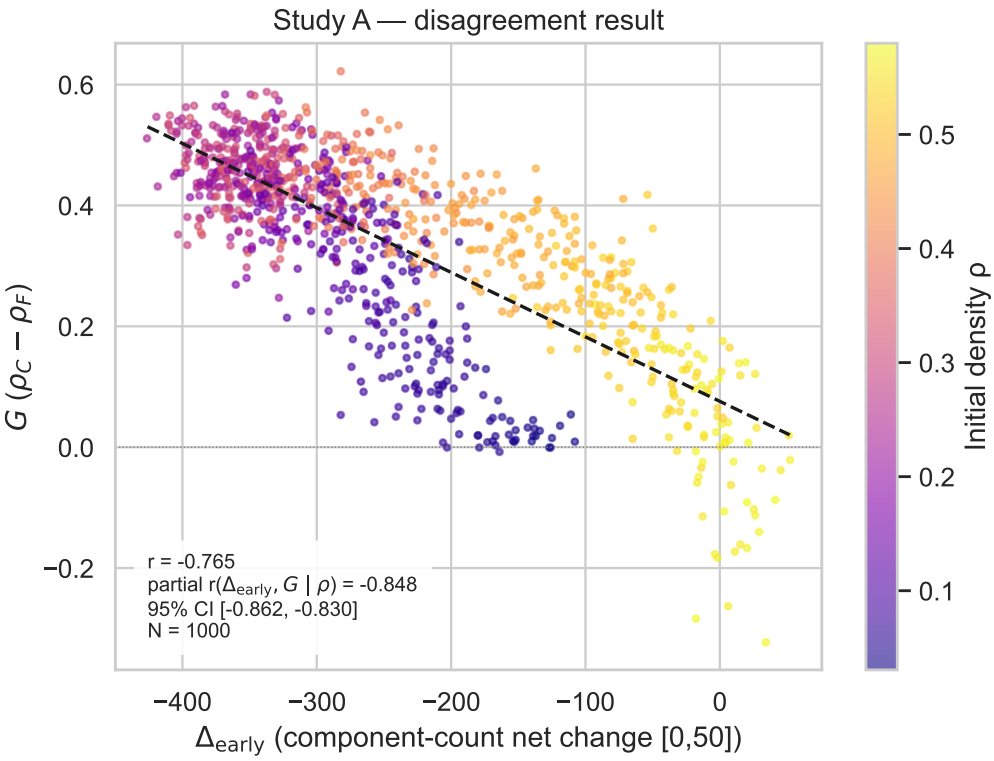

FIG. 1. Observer-disagreement main result. Early fine-scale component net change Δ_{early} predicts the full-run narrative gap G. Points are worlds coloured by initial density. The relationship sharpens after density control: raw $r = -0.765$, partial $r(\Delta_{\text{early}}, G \mid \rho) = -0.848$, 95% bootstrap CI $[-0.862, -0.830]$.

These traces are descriptive and carry no inferential weight; the quantitative result rests on the ensemble statistics above.

IV. PREDICTIVE SCALE IS TARGET-RELATIVE

A. Setup

In a 500-world 128×128 ensemble, density uniform on $[0.20, 0.35]$, we test whether the observation scale that best predicts a future observable depends on which observable one chooses. The question has a direct scientific meaning: if there is a single universally optimal scale for prediction, then scale choice affects only precision, not structure. If different future quantities prefer different scales, the implication is stronger: the choice of scale is not merely

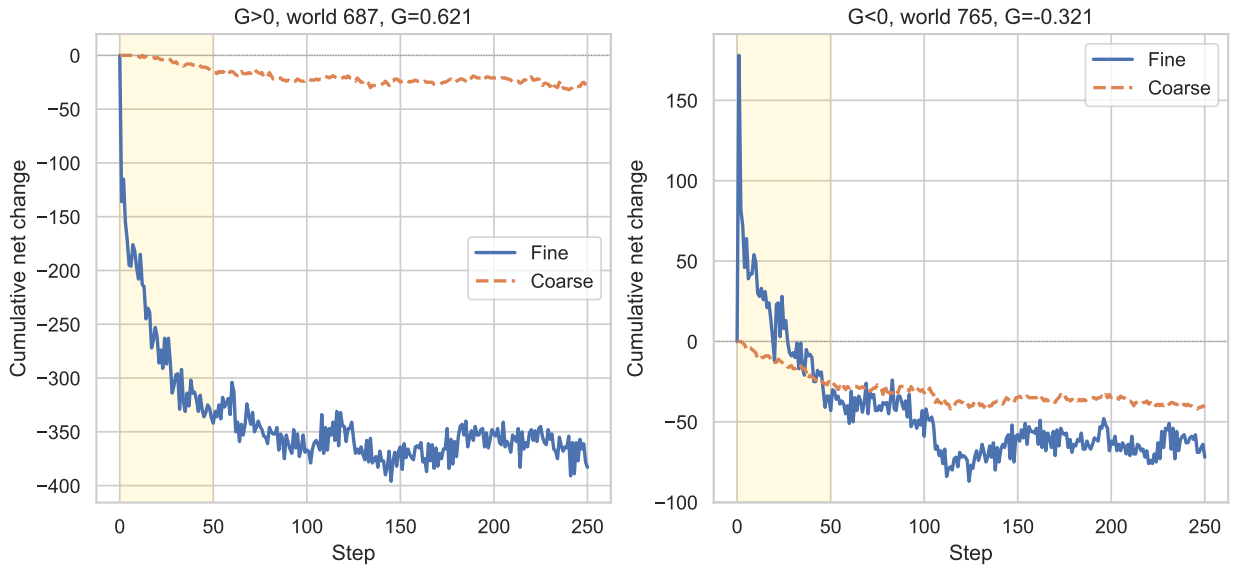

FIG. 2. Illustrative observer trajectories for a representative positive-gap world (left, $G = 0.621$) and a representative negative-gap world (right, $G = -0.321$). Fine and coarse observers can accumulate opposite net-change narratives over the same deterministic run. Yellow shading marks the early window $[0, 50]$ used to compute Δ_{early}. These panels are descriptive only and are not used in the inferential test.

a resolution question but a question about which aspects of the post-transient state are relevant to which aspects of the future.

Features are extracted at five scales $B \in \{1, 2, 4, 8, 16\}$, where at $B = 1$ the feature time series uses BFS component counts (the fine observer defined above) and at $B > 1$ it uses occupied-block counts. The predictor window is $[50, 100]$; the target window is $[100, 200]$. The two **primary targets** are $\bar{N}_{\text{live}}$ (time-averaged live-cell count) and $\bar{N}_{\text{occ}}^{B=8}$ (time-averaged occupied-$B = 8$-block count). Two **secondary targets** are $\bar{A}$ (mean step-to-step absolute change in live cells) and G_{future} (narrative gap $G[100, 200]$); secondary targets are reported but do not gate the main conclusion. The comparison varies the scale at which predictor features are extracted, while the target definitions themselves remain fixed.

Before analysis, we specified that evidence for target-relative predictive scale would require non-overlapping bootstrap peak-B intervals for at least two primary targets, with one peak at $B \leq 2$ and another at $B \geq 4$. The negative controls are static IC features and early-transient $[0, 50]$ features, expected to yield no positive held-out R^2 for primary targets.

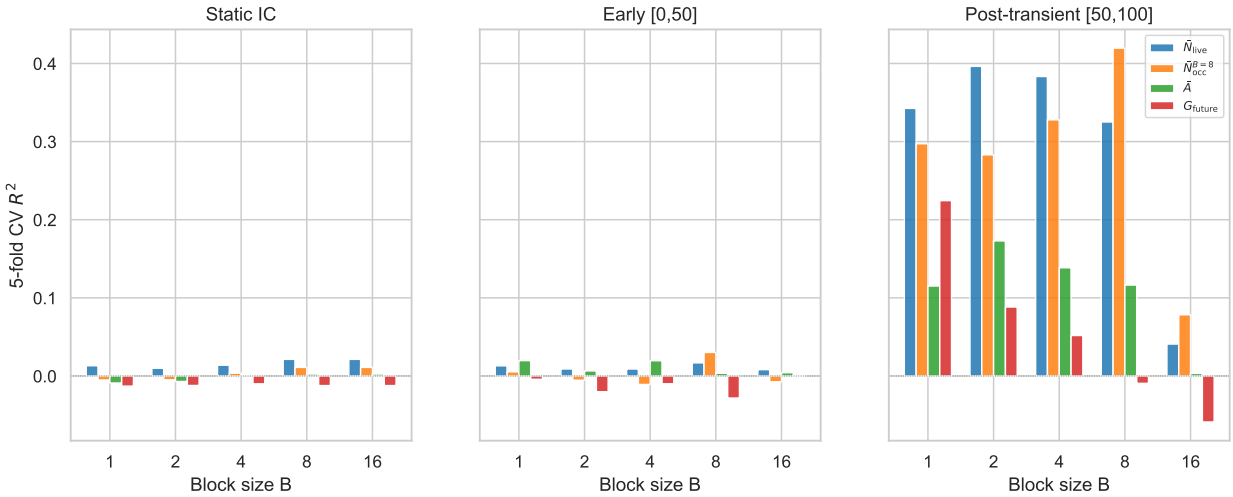

FIG. 3. Predictive signal across temporal windows. Static initial-condition features and early-transient features remain weak for the primary targets, while post-transient features carry substantial predictive power. This shows the clean negative-control result.

B. The negative controls are clean

Of the 20 primary-target static-or-early conditions, none exceeds $R^2 = 0.04$. This establishes that the predictive signal is not inherited from the initial condition or from the statistics of the early die-off. The information that forecasts the future emerges only after the initial transient has settled and the surviving spatial structure has largely stabilised. In a deterministic system the initial condition fully specifies all future states, but low-dimensional statistical summaries of that initial condition—at any scale—cannot access the relevant degrees of freedom for forecasting.

C. Post-transient predictive structure

The post-transient window contains the predictive signal. For the primary targets, $\bar{N}_{\text{live}}$ reaches cross-validated $R^2 = 0.396$ at $B = 2$, while $\bar{N}_{\text{occ}}^{B=8}$ reaches $R^2 = 0.420$ at $B = 8$ (Table III). The negative-control audit is clean: 0/20 primary-target static-or-early conditions exceed $R^2 = 0.04$. This rules out the possibility that the result is a trivial echo of the initial condition or of the earliest transient.

Table IV gives the bootstrap peak-B analysis. The confidence interval for $\bar{N}_{\text{live}}$ is $[2, 4]$;

TABLE III. Post-transient $[50, 100] \to [100, 200]$ cross-validated R^2 by target and observation scale. Bold entries mark the peak scale per primary target.

Target	$B = 1$	$B = 2$	$B = 4$	$B = 8$	$B = 16$
$\bar{N}_{\text{live}}$ (primary)	0.343	**0.396**	0.383	0.325	0.041
$\bar{N}_{\text{occ}}^{B=8}$ (primary)	0.297	0.283	0.328	**0.420**	0.078
$\bar{A}$ (secondary)	0.115	0.173	0.139	0.116	0.003
G_{future} (secondary)	0.224	0.088	0.052	−0.009	−0.059

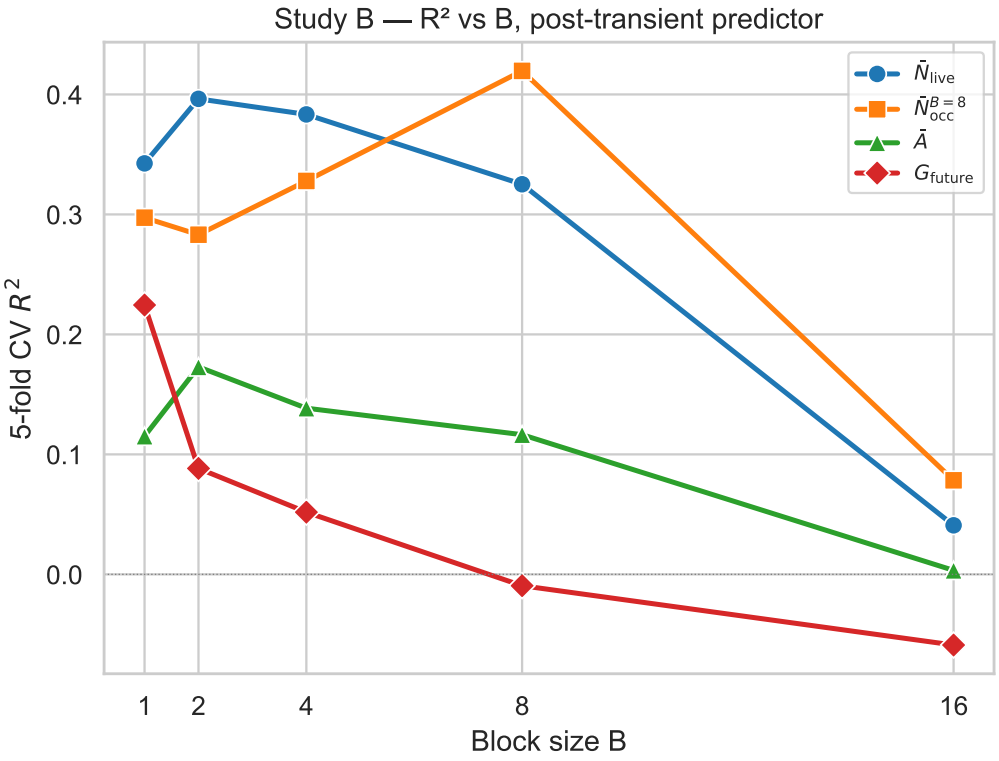

FIG. 4. Target-relative predictive scales. Future live-cell count peaks at $B = 2$, while future occupied-block count peaks at $B = 8$. The two primary targets prefer different observation scales within the same deterministic system.

the interval for $\bar{N}_{\text{occ}}^{B=8}$ is $[8, 8]$—meaning not one of the 1000 bootstrap draws placed the peak anywhere other than $B = 8$. The two intervals do not overlap, and the required pattern (one peak at $B \leq 2$, another at $B \geq 4$) is present.

TABLE IV. Bootstrap peak-B confidence intervals for primary targets (1000 resamples, world-row resampling).

Target	Point peak B	95% CI
$\bar{N}_{\text{live}}$	2	[2, 4]
$\bar{N}_{\text{occ}}^{B=8}$	8	[8, 8]
CI overlap	None	
Required pattern ($B \leq 2$ and $B \geq 4$ among primaries)	Present	
Negative control: primary-target static/early with $R^2 > 0.04$	0/20	

D. Interpretation

This provides direct evidence, within this system, against the existence of a single privileged predictive scale. The scale preferences have a qualitative logic. $\bar{N}_{\text{live}}$ benefits from a smoothing scale ($B = 2$) that attenuates single-component fluctuations while preserving the spatial heterogeneity of surviving structures; neither the finest scale nor the coarse-block scale achieves this trade-off as well. $\bar{N}_{\text{occ}}^{B=8}$ is best predicted at its own native scale ($B = 8$): the coarse observer's own dynamics are maximally informative about its own future population, a form of internal self-consistency that does not generalise to other observables. The secondary target G_{future} peaks at $B = 1$, consistent with Section III: the observer gap is driven by fine-scale events invisible to coarser descriptions. These are different questions about the same deterministic system, and they are best answered from different vantage points.

The two primary targets' bootstrap peak-B confidence intervals do not overlap: $\bar{N}_{\text{live}}$ peaks at $B = 2$ with CI $[2, 4]$, while $\bar{N}_{\text{occ}}^{B=8}$ peaks at $B = 8$ with CI $[8, 8]$. The exact $[8, 8]$ interval for $\bar{N}_{\text{occ}}^{B=8}$ is especially strong: in the bootstrap resampling, no alternative scale displaced the block observer's own native scale as the best predictor of its future occupied-block population.

V. A STABLE LAW FOR COMPONENT CHANGE

A. Motivation for the embedded-isolated-cell predictor

In the post-transient GoL regime, the grid contains a mixture of stable structures—still lifes, oscillators, and occasional gliders—interspersed with near-isolated cells that are alive but structurally marginal. An embedded cell is alive, has no 4-connected live neighbours (so it is not part of any larger component), but does have at least one diagonal live neighbour. This structural configuration makes such cells natural candidates as predictors. The analysis below tests whether their count at the start of a measurement window statistically predicts how much the component count will change over that window.

B. Design and criterion

The analysis covers six conditions: GoL on 64^2 and 128^2 grids, each at low ($\rho \in [0.10, 0.20]$), mid ($[0.25, 0.35]$), and high ($[0.35, 0.50]$) density bands, one fixed seed per condition. In each condition we fit

$$\text{fine_net} \sim \beta_{\text{emp}} \cdot \text{iso_emb} + c(\text{area}) \tag{3}$$

by OLS, where $\text{fine_net} = \text{net}_F[T_0, T_1]$ and iso_emb is the embedded isolated cell count at T_0. The criterion, specified in advance, is $\text{CV}(\beta_{\text{emp}}) \leq 0.20$ across the six conditions.

C. Results

Table V reports the slope estimates.

The slope varies between -1.38 and -1.77 across grid sizes and density regimes. The mean slope is -1.52, with a standard deviation of 0.13 and CV $=$ 0.087. This is well below the threshold, establishing

$$\text{fine_net} \approx -1.52 \cdot \text{iso_emb} + c(\text{area}) \qquad \text{CV} = 0.087, \quad \text{six GoL conditions} \tag{4}$$

as a stable empirical law across these conditions. Under the component-count fine observer used throughout this paper, the mean slope is approximately -1.52 units of future component net change per embedded isolated cell.

TABLE V. OLS slope estimates across the six GoL conditions. The CV across conditions is well below the threshold of 0.20.

Condition	β_{emp}	R^2
64^2, low density	-1.462	0.413
64^2, mid density	-1.613	0.338
64^2, high density	-1.383	0.296
128^2, low density	-1.767	0.684
128^2, mid density	-1.404	0.302
128^2, high density	-1.498	0.328
Mean $\pm$ SD	-1.521 ± 0.133	
CV	0.087	

D. Interpretation

The law is empirical, not mechanistic. It says that embedded isolated cells predict future component-level decline with a coefficient stable across size and density. It does not say why, and Section VI will show that the simplest mechanistic account—that the law is driven by a positive residual coupling over and above trivial null dynamics—is not supported. The distinction matters: the law is as solid as the CV figure suggests; the mechanism behind it is an open question.

The R^2 values vary more than the slopes do, ranging from 0.30 to 0.68. This variation reflects the different dynamical character of the density regimes rather than instability in the relationship: in low-density conditions the future trajectory is more structured and thus easier to predict from a single feature, while in high-density conditions the dynamics are more variable.

E. Circularity and selection concerns

An obvious concern is circularity: a predictor can appear powerful if it is defined using the very future behaviour it is later said to predict. Our construction avoids this. Embedded isolated cells are identified entirely from the configuration at the observation time T_0, using

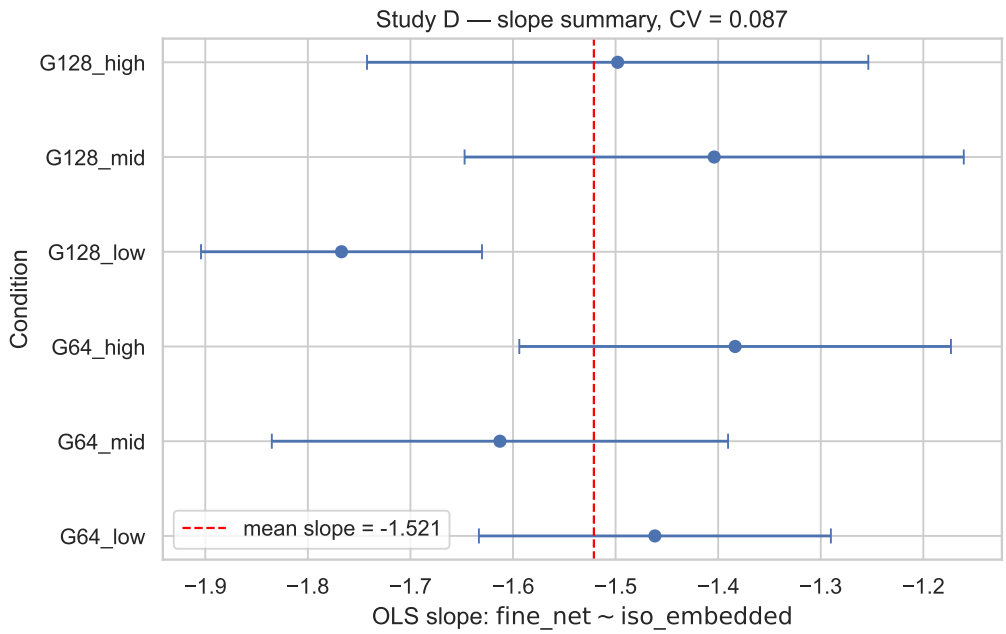

FIG. 5. Slope estimates across the six GoL conditions. Points show OLS slopes and horizontal bars show 95% confidence intervals. The mean slope is approximately -1.52 and the cross-condition coefficient of variation is 0.087, below the pre-specified stability threshold.

a contemporaneous structural criterion (Section II): alive, no 4-connected live neighbours, at least one diagonal live neighbour. The outcome variable fine_net—net fine-scale component change over the subsequent window—is measured afterward and plays no role in the object definition. The relation is therefore temporally prospective, not definitional.

A second concern is that the predictor might re-express current fragmentation, making the law a bookkeeping identity rather than a predictive regularity. But the target is future change in component count, not the present count itself, and the empirical slope is a cross-condition regularity whose stability (CV = 0.087) would not be expected from an accounting tautology. The mediation and null-decomposition analyses in Section VI further confirm that the relationship carries non-trivial structure.

A third concern is selection: were many candidate predictors tried, with only the successful one reported? The embedded isolated cell is the structurally motivated candidate—it is the only cell-level class that is simultaneously 4-disconnected (hence a singleton component) and 8-connected to an existing structure (hence embedded rather than truly alone). No automated search over predictor families was conducted; the comparison reported in Sec-

tion VI between embedded and alone cells is the only competitor tested and is presented in full.

VI. TESTS OF STRONGER MECHANISM INTERPRETATIONS

The preceding sections establish three empirical regularities. This section tests two stronger mechanism interpretations: first, whether embedded isolated cells mediate the path from early structure to observer disagreement; second, whether the empirical law survives as a positive excess beyond death and local-neighbourhood nulls. Neither interpretation is supported by the present analyses.

A. Global mediation

The proposed mechanism is that early accumulation of embedded isolated cells drives future component net change, which in turn drives the narrative gap. We test this by the product-of-coefficients method, controlling for density in both the mediator and outcome regressions, with a 95% CI from 1000 bootstrap resamples.

Embedded count predicts G more strongly than alone count ($r = 0.471$, partial $r = 0.632$ vs. $r = 0.328$, partial $r = 0.403$), and embedded count strongly predicts fine_net ($r = -0.731$, partial $r = -0.608$). These comparisons show that embedded isolated cells are the stronger discriminator.

However, the mediation test fails. The estimated indirect effect is -4.61×10^{-4}, with mediation fraction -0.072 and bootstrap 95% CI $[-0.145, -0.007]$. This interval lies entirely below zero and fails to exceed 0.50. The supported reading is therefore that embedded isolated cells are harbingers of later disagreement, not established global mediators of it.

B. Null decomposition of the component-change law

We decompose $\beta_{\text{emp}} = \beta_{\text{death}} + \beta_{\text{local}} + \beta_{\text{residual}}$ and ask whether β_{residual} represents a positive excess beyond trivial null dynamics. The death term $\beta_{\text{death}} = -1.0$ is analytically fixed: each embedded cell that dies removes exactly one component. A static local null (no neighbourhood context) gives $\beta_{\text{local}} \approx 0$, leaving $\beta_{\text{residual}} \approx \beta_{\text{emp}} - \beta_{\text{death}}$; the excess index

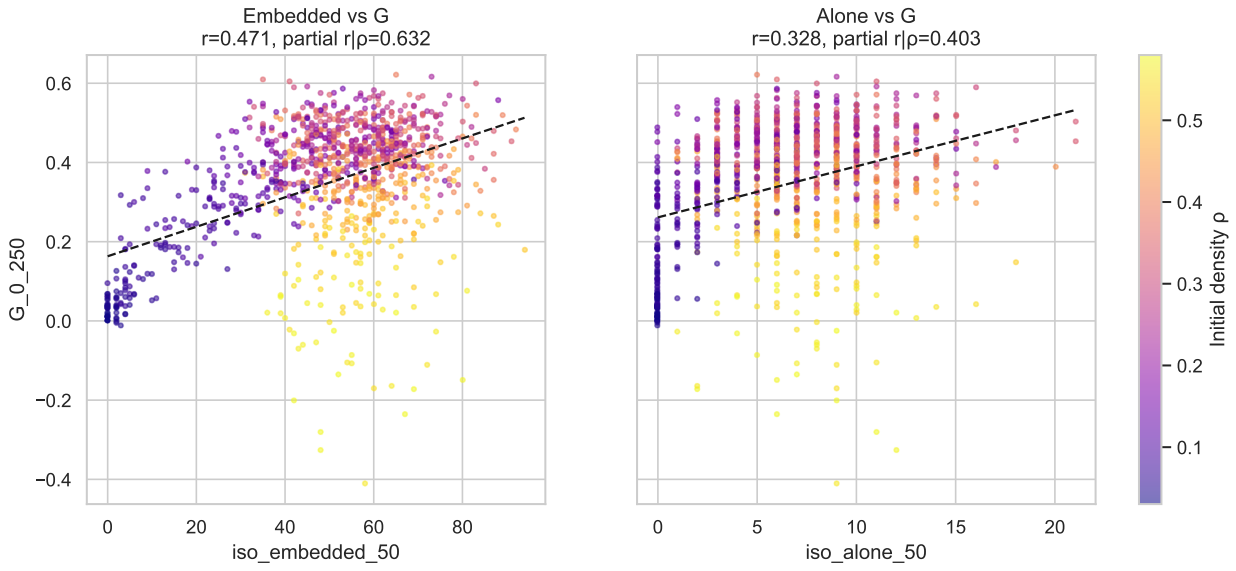

FIG. 6. Association between isolated cell count at $t = 50$ and narrative gap G for embedded cells (left) and alone cells (right). Embedded cells are the stronger discriminator ($r = 0.471$, partial $r = 0.632$ vs $r = 0.328$, partial $r = 0.403$ for alone cells). The mediation test asks whether this association runs through future component net change; it does not meet the pre-specified threshold.

$\chi = \beta_{\text{residual}}/|\beta_{\text{death}} + \beta_{\text{local}}|$ is negative in all six conditions ($\chi \in [-0.38, -0.77]$), so even under the static null the residual is not positive.

A dynamic local null, using 5×5 neighbourhood patches centred on each focal cell, gives $\beta_{\text{local}} \in [0.35, 0.51]$ across conditions. Here β_{local} is an accounting term defined from mean single-step local structural contributions, not a regression coefficient from a separate null-model fit. Local neighbourhood context is itself predictive and non-negligible. Under this null, $\beta_{\text{residual}} \in [-0.81, -1.12]$ and $\chi_{\text{mean}} = -1.78$; no condition satisfies $\chi \geq 0.25$. The residual remains large and negative: not a small positive excess over local dynamics, but a large negative residual after subtracting the death and local terms.

The empirical law is stable (CV = 0.087). The claim that it arises from a positive residual coupling beyond death and local neighbourhood dynamics is not supported.

VII. DISCUSSION

a. Supported results. Three empirical regularities in GoL are robust to fixed observer definitions and fixed decision thresholds. Observer disagreement is strongly predictable from

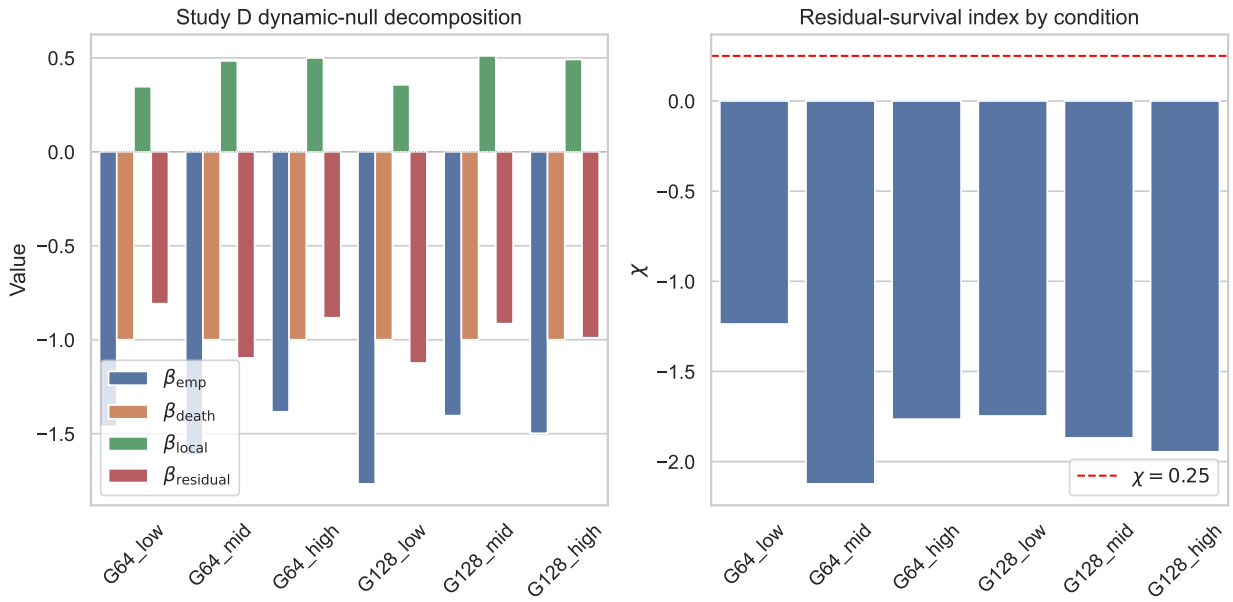

FIG. 7. Dynamic-null decomposition of the component-change law. Across conditions, the empirical slope is stable, but the residual term does not survive as a positive excess beyond death and local-neighbourhood contributions. The empirical law is therefore supported; the stronger residual-coupling interpretation is not.

early fine-scale component dynamics: the partial correlation between early component net change and the full-run narrative gap is $r = -0.848$ and sharpens after density control. The scale that best forecasts a future quantity depends on which quantity: future live-cell count peaks at $B = 2$, future occupied-block count peaks at $B = 8$, and the two bootstrap confidence intervals do not overlap. The embedded-isolate slope is stable across six size-by-density conditions with CV $= 0.087$. Together these results show that the choice of coarse-grained description in GoL is not arbitrary: different measurement schemes yield different but lawful summaries of change, and their disagreement is itself predictable from early local structure.

b. Unsupported mechanism interpretations. Two mechanism interpretations are not supported. Embedded isolated cells are harbingers of later observer disagreement, but not established global mediators of it: the product-of-coefficients mediation fraction is -0.072 with CI entirely below zero. The empirical slope law does not emerge as a positive residual over the dynamic local null: $\chi < 0$ in all six conditions. The empirical structure is well-characterised; the mechanistic account is not yet in hand.

c. Observer definition and numerical scope. All analyses in this paper use a single fine measurement scheme: 4-connected component count on the torus. The predictor—embedded isolated cell count—is defined from present-time structure only and does not use future trajectory information; the law is prospective, not circular (Section V). Under this scheme the mean slope is approximately -1.52 units of future component net change per embedded isolated cell. The slope is design-dependent: a narrower-scope study restricted to $\rho \in [0.20, 0.35]$ with a 100-step future window yields $\beta \approx -1.27$ with lower variance (CV $= 0.063$). The sign and order of magnitude are stable across tested designs; the functional dependence $\beta(\rho, K, \text{rule set})$ is uncharacterised. Because component counts and live-cell counts behave differently, this coefficient should not be compared directly to live-cell-based estimates.

d. Open problems. The mechanism question is the obvious open problem. The global mediation path through fine_net is insufficient; a more structured or non-global path may fare better. The dynamic local null leaves a large negative β_{residual}, which indicates the slope law is driven by something beyond isolated-cell deaths and immediate neighbourhood context—but what that is remains uncharacterised. Whether any of these results generalise beyond GoL, beyond the density ranges tested, or to other Life-like rules is a question this paper does not answer.

ACKNOWLEDGMENTS

No external funding was received for this work. The author has no conflicts of interest to declare. Simulations were performed using Python 3 with NumPy, SciPy, scikit-learn, pandas, Matplotlib, and seaborn.

DATA AND CODE AVAILABILITY

All code, simulation outputs, and figure sources are available at https://github.com/kunalb541/CA.

REFERENCES

[1]Crutchfield, J. P. and Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2):105–108.

[2]Flack, J. C. (2017). Coarse-graining as a downward causation mechanism. *Phil. Trans. R. Soc. A*, 375(2109):20160338.

[3]Gardner, M. (1970). Mathematical games: The fantastic combinations of John Conway's new solitaire game "life". *Scientific American*, 223(4):120–123.

[4]Hoel, E. P., Albantakis, L., and Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proc. Natl. Acad. Sci.*, 110(49):19790–19795.

[5]Israeli, N. and Goldenfeld, N. (2004). Computational irreducibility and the predictability of complex physical systems. *Phys. Rev. Lett.*, 92(7):074105.

[6]Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3):12–37.

[7]Rupe, A. and Crutchfield, J. P. (2018). Local causal states and discrete coherent structures. *Chaos*, 28(7):075312.

[8]Schreiber, T. (2000). Measuring information transfer. *Physical Review Letters*, 85(2):461–464.

[9]Shalizi, C. R. and Moore, C. (2003). What is a macrostate? Subjective observations and objective dynamics. *arXiv preprint cond-mat/0303625*.

[10]Tononi, G., Sporns, O., and Edelman, G. M. (1994). A measure for brain complexity: Relating functional segregation and integration in the nervous system. *Proc. Natl. Acad. Sci.*, 91(11):5033–5037.

[11]Vichniac, G. Y. (1984). Simulating physics with cellular automata. *Physica D: Nonlinear Phenomena*, 10(1–2):96–116.

[12]Wolfram, S. (1984). Universality and complexity in cellular automata. *Physica D: Nonlinear Phenomena*, 10(1–2):1–35.